

Boyce-Codd Normal Form (BCNF) — Relational Database Design

Definition: A relation is in **Boyce-Codd Normal Form (BCNF)** (also known as 3.5NF) if and only if for **every non-trivial functional dependency** $X \rightarrow Y$, X is a strict Super Key.

1. Detailed Technical Explanation

BCNF is a stricter version of 3NF that handles cases where a relation has **multiple overlapping candidate keys**.

3NF vs BCNF Comparison:

Property	3NF	BCNF
Allowed Condition $X \rightarrow Y$	X is Super Key OR Y is Prime Attribute	X MUST be a Super Key (No exceptions!)
Dependency Preservation	Always Guaranteed	Not always guaranteed
Redundancy	Allows minor non-key redundancy	Completely eliminates functional redundancy

Conversion Example to BCNF

Relation ADVISOR(Student_ID, Subject, Advisor_Name)

- **Assumptions:**
 1. A student can choose multiple subjects.
 2. For each subject, a student is assigned one advisor.
 3. Each advisor advises only ONE subject.
- **Candidate Keys:** {Student_ID, Subject} and {Student_ID, Advisor_Name}
- **Functional Dependencies:**
 1. {Student_ID, Subject} $\rightarrow$ Advisor_Name
 2. Advisor_Name $\rightarrow$ Subject (Violation! Advisor_Name determines Subject, but Advisor_Name is NOT a super key)

Relation Status:

- Is it 3NF? **YES**, because Subject is a prime attribute.
- Is it BCNF? **NO**, because in $\text{Advisor_Name} \rightarrow \text{Subject}$, **Advisor_Name** is NOT a super key.

BCNF Decomposition:

1. **ADVISOR_SUBJECT (Advisor_Name, Subject)**
 - Primary Key: **Advisor_Name**
2. **STUDENT_ADVISOR (Student_ID, Advisor_Name)**
 - Primary Key: {Student_ID, Advisor_Name}

2. Core Concepts & Memory Keywords

- **Strict Super Key Rule:** Every determinant in a functional dependency must be a super key.
 - **Overlapping Candidate Keys:** Candidate keys sharing common attributes.
 - **3.5 Normal Form:** Stronger normalization standard than 3NF.
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3. Must-Write Points for Exams

- BCNF eliminates all redundancy resulting from functional dependencies.
 - Every relation in BCNF is guaranteed to be in 3NF, 2NF, and 1NF.
 - Decomposing a 3NF relation into BCNF may sometimes sacrifice **dependency preservation**.
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4. Quick Recall Flow

3NF Relation -> Inspect all FDs ($X \rightarrow Y$) -> Verify if X is Super Key -> Decompose non-superkey determinants -> BCNF Achieved